

Bachu Vaibhavi

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Summary

AI/ML Engineer with experience building production-ready AI, Machine Learning, NLP, and Generative AI solutions using Python, SQL, LLMs, Prompt Engineering, and cloud technologies. Skilled in developing scalable AI applications, agentic workflows, intelligent automation systems, and conversational AI solutions that drive measurable business impact.

Skills

- **Languages & Development:** Python, SQL, Git, GitHub, REST APIs, Backend Development
- **AI/ML & NLP:** Machine Learning, Natural Language Processing (NLP), Scikit-Learn, TensorFlow, PyTorch, Feature Engineering, Model Evaluation
- **LLMs & Prompt Engineering:** Prompt Engineering, RAG, AI Agents, LangChain, LlamaIndex, OpenAI GPT, Claude, Gemini, Vector Databases
- **Cloud & Automation:** N8N, GitHub Actions, Claude Agents, Claude Code, Tableau, Power BI, Docker, MinIO, CI/CD Concepts
- **Cloud & DevOps:** Docker, Linux, MinIO, CI/CD Concepts, Cloud Deployment

Education

CMR Institute of Technology, CGPA : 8.25

Bachelor's in Computer Science and Engineering

Nov 2022 – Aug 2026

Work Experience

AI/ML Engineer | Sensovibe Reliability Pvt. Ltd.

Feb 2026 – Present

- Developed an AI-powered vibration monitoring and recommendation dashboard using Python, Machine Learning, and predictive analytics, enabling maintenance teams to identify potential equipment failures early and reducing unplanned **downtime by 25%**.
- Designed and implemented intelligent fault-detection models for vibration data analysis, improving anomaly detection **accuracy by 30%** and supporting faster root-cause analysis for industrial assets.
- Built AI agents and workflow automation solutions to streamline repetitive operational tasks, **reducing manual effort by 60%** and improving overall team productivity.
- Built LLM-powered chatbots using NLP and Prompt Engineering, reducing troubleshooting **response time by 70%**.
- Automated data processing, reporting, and alert-generation workflows using Python, GitHub Actions, and AI-based automation pipelines, **decreasing report preparation time by 80%** and improving operational efficiency.
- Collaborated with cross-functional engineering teams in a fast-paced startup environment to design, deploy, and optimize scalable AI solutions, **contributing to a 40% faster delivery cycle** for new product features and customer requirements.

Projects

Intelligent Chatbot with NLP & Prompt Engineering for Vibration Monitoring System

- Developed an AI-powered industrial maintenance chatbot using Python, NLP, Prompt Engineering, and RAG to analyze vibration monitoring data, fault histories, and maintenance records.
- Designed agentic workflows using Claude Agents and LangChain to provide real-time fault diagnosis and troubleshooting recommendations, **reducing manual analysis effort by 65%**.
- Implemented contextual retrieval and prompt optimization techniques that improved response accuracy by 32% and reduced **troubleshooting time by 40%**.

AI-Powered Trading Intelligence Platform

- Built a machine learning-based stock prediction platform using Python, XGBoost, LSTM, and financial market indicators to forecast short-term price movements.
- Engineered automated data pipelines for data collection, preprocessing, feature generation, and model retraining, improving prediction accuracy by **18% compared to baseline models**.
- Developed an AI-driven insights dashboard that **reduced manual market analysis time by 55%** and enabled faster investment decision-making.

Certificates

SQL Certificate - Udemy | Oracle Analytics Cloud - Oracle | Prompt Engineering - AWS